

Saba Amreen Mohammad

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Healthcare Data Analyst | SQL | Python | Power BI | Machine Learning & Predictive Analytics

SUMMARY:

Healthcare Data Analyst with 3+ years of experience leveraging SQL, Python, and Power BI to analyze EHR, claims, and clinical datasets. Proven expertise in building scalable ETL pipelines, performing advanced analytics, and developing machine learning models to improve patient outcomes and operational efficiency. Strong background in healthcare standards, including FHIR, ICD-10, and HIPAA compliance, with a track record of delivering actionable insights through exploratory data analysis (EDA), data wrangling, and stakeholder reporting.

KEY EXPERTISE:

Healthcare Analytics, Predictive Modeling, ETL Pipelines, Data Visualization, Clinical Data Analysis, Claims Data Analysis

TECHNICAL SKILLS:

Programming: Python (Pandas, NumPy, scikit-learn), SQL (Joins, Window Functions)

Visualization: Power BI, Tableau, Excel (Power Query, Pivot Tables)

Machine Learning: Regression, Classification, Clustering, Model Evaluation (ROC-AUC, Precision-Recall)

Databases & Cloud: MySQL, SQL Server, Snowflake, BigQuery

Healthcare Data: EHR, Claims, FHIR, ICD-10, HEDIS, MIPS

Other: Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), Feature Engineering, ETL Pipelines, Data Validation, KPI Reporting, Stakeholder Communication

PROJECTS:

Patient Readmission Prediction Model

- Built machine learning models using Python (scikit-learn) to predict patient readmission risk from healthcare datasets
- Performed feature engineering and data preprocessing on structured clinical data
- Conducted exploratory data analysis (EDA) to identify patterns and trends in healthcare datasets
- Evaluated model performance using ROC-AUC and classification metrics

Claims Data Analysis

- Analyzed claims datasets using SQL (joins and window functions) to identify denial trends and root causes
- Developed Power BI dashboards to visualize claims patterns and financial impact
- Generated insights and reports to support business decision-making and KPI tracking

EXPERIENCE:

Healthcare Data Analyst | AbbVie, Waukegan, IL

Sep 2025 – Present

Leveraged EHR and claims datasets to develop analytics solutions and predictive models supporting clinical and operational decision-making.

- Designed and deployed Power BI dashboards integrating EHR and claims data to track healthcare quality metrics, reducing reporting turnaround time by ~25%
- Implemented and optimized ETL pipelines using Python (Pandas) and SQL to integrate EHR and claims data, improving data processing efficiency by ~30% and reducing data inconsistencies
- Developed machine learning models (logistic regression, decision trees) to identify high-risk patient populations, improving early intervention targeting by ~18%
- Performed data cleaning, data wrangling, and exploratory data analysis (EDA) to ensure data quality and support analytical insights
- Automated CMS quality reporting and readmission analysis workflows using Python and Power BI, reducing manual effort by ~30%
- Collaborated with compliance and governance teams to ensure HIPAA- and GDPR-compliant data workflows

Healthcare Data Analyst | Change Healthcare, Nashville, TN

Dec 2024 – Aug 2025

Delivered data analysis solutions to improve claims processing, reimbursement workflows, and population health insights.

- Developed complex SQL queries using joins, aggregations, and window functions to extract, analyze, and validate claims data, improving query performance by ~25%
- Built Power BI dashboards to monitor reimbursement cycles and operational KPIs, improving reporting efficiency by ~25%
- Engineered ETL pipelines integrating EHR and FHIR datasets using Python and SQL to improve data accessibility and interoperability
- Automated PHI de-identification and validation processes using Python to support secure analytics and research
- Collaborated with IT teams to implement role-based access control and audit mechanisms aligned with HIPAA requirements

Clinical Data Coordinator | PrimEra Medical Technologies, India

Oct 2021 – Jul 2023

Managed clinical data workflows supporting regulatory reporting and healthcare analytics.

- Integrated clinical data from EHR, pharmacy, and laboratory systems to create unified datasets for analysis and reporting
- Prepared CDISC-compliant datasets (SDTM, ADaM) and generated SAS-based TLF reports for regulatory submissions
- Developed Tableau dashboards and reports to track KPIs and support data-driven decision-making
- Performed data validation and quality checks, reducing data discrepancies by ~20% to ensure compliance with GCP and HIPAA standards. Supported data transformation and preprocessing for clinical research datasets

EDUCATION:

Master of Science in Health Data Science, Saint Louis University School of Medicine

Aug 2023 – Aug 2025

Doctor of Pharmacy (PharmD), Jawaharlal Nehru Technological University, India

Sep 2017 – Jul 2023